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The remagnetization of Devono-Carboniferous sediments from the Ardenno-Rhenish Massif

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ABSTRACT

A paleomagnetic study was conducted on the Paleozoic sedimentary rocks of the Ardenno-Rhenish Massif and the Brabant Massif. Lower Carboniferous and Middle Devonian carbonates in the Dinant Synclinorium and Inde Syncline carry a thermoviscous remagnetization residing in magnetite. Lower Devonian red beds in the Dinant Synclinorium carry a chemoremanent remagnetization residing in hematite. The sediments were remagnetized by burial metamorphism during the Asturian phase of the Variscan orogeny. With a few exceptions, in all formations the magnetizations are of inverse polarity, corresponding to the Kiaman paleofield in the Permo-Carboniferous.

The virtual geomagnetic poles (VGPs) of the Inde Syncline agree with the Permo-Carboniferous pole position of Europe. VGPs of the Dinant Synclinorium which deviate from the Permo-Carboniferous reference direction are due to syntectonic remagnetizations and subsequent 10–20° (maximum 40°) dextral shear deformations. These deformations can be correlated with the northwards overthrusting of the Dinant Nappe. For the Inde Syncline post-Asturian rotational deformations can be excluded. Should a N–S transport of this part of the Dinant Nappe not have occurred, the declination pattern indicates that the western part of the Dinant Nappe was transported further north than the eastern part.

Introduction

For two main reasons the Paleozoic rocks of the Ardenno-Rhenish Massif and the Brabant Massif (Fig. 1) have to a large extent not seen any paleomagnetic research: firstly, the intense rock deformation impedes the reconstruction of pretectonic magnetic directions and no general method exists to correct internal rock deformation (Lowrie et al., 1986; Cogné and Perroud, 1987); secondly, in most cases primary magnetizations often have been completely overprinted by the Variscan remagnetization (McCabe and Elmore, 1989).

Paleomagnetic data are of special significance for the reconstruction of Variscan Central Europe. Geotectonic models of the Variscides may be placed into two groups, those based predominantly on paleomagnetic data and those based partly on such data. In the first group are those developed by Perroud et al. (1984), Perroud (1986) and Van der Voo (1990), who have concluded, from concordant paleomagnetic directions of various peri-Atlantic microblocks, the integrity of a Paleozoic "Armorica Plate", which may possibly include the Central European Variscides. In the second group of models are those of Lorenz and Nicholls (1984), which assume that the southern border of the Rhenohercynian Zone results from the collision of Gondwana and Laurasia, and those by André et al. (1986), which assume a subduction zone at the southern border of the Brabant Massif.

As yet, paleomagnetic data from the Rhenohercynian Zone [Harz—Rother (1971), Bachtadse (1984); Sauerland—Bachtadse (1984); Eifel —Nowaczyk and Bleil (1985); Ardennes—Edel and Coulon (1987)] have not been able to provide a paleogeographical reconstruction of this part of the central European Variscides due to remagnetization. In this study, a systematic paleomagnetic and rock-magnetic investigation of the Paleozoic sedimentary rocks from the Ardenno-Rhenish Massif and the Brabant Massif have been carried

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out and the results are interpreted in a magnetotectonic analysis.

Geological setting, sampling and measurements

Geological setting

The Ardenno-Rhenish Massif belongs to the Rhenohercynian Zone and is located near the northern front of the European Variscan orogenic belt (Fig. 1). The Caledonian Ardennes massifs and the Brabant Massif are also part of the Variscan orogen. The main structural units, which have been investigated paleomagnetically, are the Caledonian Brabant and Ardennes massifs and the Variscan Dinant–Namur Synclinorium and Inde Syncline (Fig. 1).

The massifs of Brabant, Stavelot–Venn and Rocroi consist of Cambrian–Silurian clastic sediments of the early Paleozoic Ardennes Basin, which have been folded in the Caledonian orogeny under low-grade metamorphic conditions (Walter, 1980). During the Devonian and Early Carboniferous, the development of the Variscan geosyncline resulted in the deposition of more than 6000 m of sediments in the Dinant–Namur Synclinorium and more than 4000 m in the Inde Syncline. In the Asturian phase the sediments of the Subvariscan marginal basin were folded and faulted. This Variscan deformation resulted in E–W- to NE–SW-trending fold structures. The Dinant Synclinorium and, in its eastern continuation, the Ardennes and the Linksrheinische Schiefergebirge have been thrusted at least 40 km to the

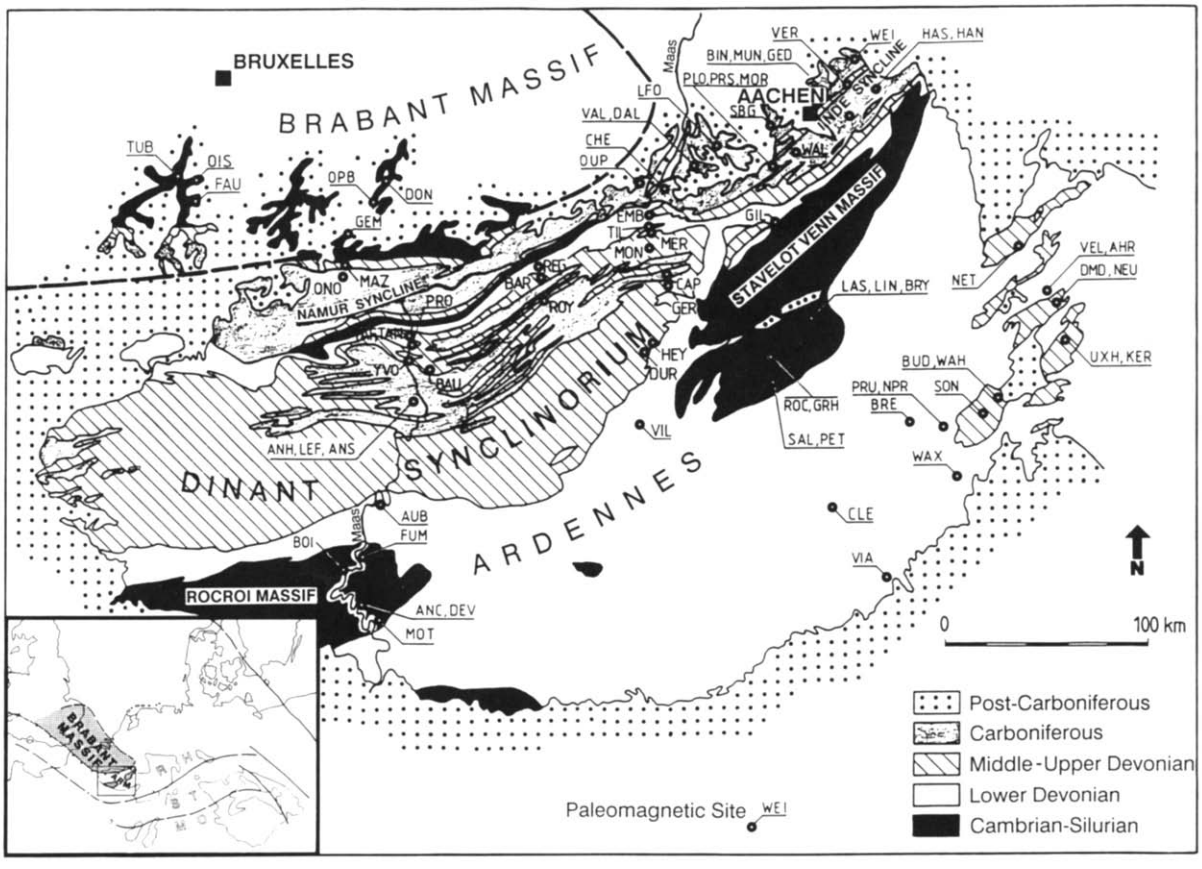


Fig. 1. Geological map of the Ardenno-Rhenish Massif (ARM), which is part of the Rhenohercynian Zone of the Central European Variscides (see inset map: *RH*, *ST*, *MO* represent Rhenohercynian, Saxothuringian, Moldanubian, respectively). Structural units which have been sampled systematically are the Variscan Dinant–Namur Synclinorium and the Inde Syncline. Paleomagnetic sampling sites are indicated.

north onto the southern flank of the Brabant Massif ("Dinant Nappe"). The overthrust zone, "Faille du Midi", can be traced westwards to the French coast and eastwards into the Aachen area (Bless et al., 1980; Meissner et al., 1981; Durst, 1985).

In the Dinant Synclinorium metamorphic age and degree of metamorphism decrease from south to north (Dandois, 1981; Pique et al., 1984). The southern border of the synclinorium is characterized by epizonal metamorphism which reached its peak in the Early Carboniferous (336 Ma). The northern border has only attained anchizonal metamorphic conditions, which took place in the Middle and Late Carboniferous (297 Ma). Cleavage and metamorphism are of the same age (Pique et al., 1984).

For the northern Linksrheinische Schiefergebirge the peak of metamorphism and the main deformation is dated at 310 Ma by means of fabric analysis and K-Ar datings (Spaeth et al., 1985; Kramm et al., 1985). The maximum metamorphic conditions prevailing south of the Stavelot–Venn Massif are estimated between 360–400°C/1–2 kbar (Kramm et al., 1985) and 307°C/2.85 kbar (Massonne, 1989). The strong decrease in foliation and illite crystallinity (Spaeth et al., 1985) as well as the lower coalification of organic material (Teichmüller and Teichmüller, 1979) in the rock sequence of the Inde Syncline indicate that the folding took place here at shallower depth than further to the south in the Stavelot–Venn Massif.

Sampling

A total of 72 outcrops were sampled in the Variscan Dinant–Namur Synclinorium, the northern Linksrheinisches Schiefergebirge (Wurm Syncline, Inde Syncline), the Eifel Synclinorium, the Eifel–Nord–Süd Zone and in the Caledonian Massifs of Brabant, Stavelot–Venn and Rocroi (Fig. 1). Preliminary paleomagnetic analyses showed that the lower Paleozoic rocks (Brabant Massif, Stavelot–Venn Massif, Rocroi Massif) are weakly magnetized and generally unsuitable for demagnetization analyses. Consequently, the systematic paleomagnetic sampling was concentrated on the Lower Carboniferous and Middle Devonian carbonates and Lower Devonian red beds in the Dinant–Namur Synclinorium and in the Inde Syncline (22 sites, Table 1).

Measurements

The magnetic measurements were made at the magnetic laboratories of the Universities of Bremen, Rennes and Köln. Three types of cryogenic magnetometers were used: the Cryogenic Consultants GM402 magnetometer (Bremen), the LETI magnetometer (Rennes) and the Superconducting Technology ScT magnetometer (Köln). Demagnetization analyses were carried out using Schonstedt GSD1 AF-demagnetizer and Schonstedt TSD1 furnaces.

A total of 735 cores have been sampled for NRM measurements. A total of 882 specimens have been analyzed by thermal, alternating field, combined thermal/alternating field (initial heating steps of 100–250°C) and chemical demagnetization. Magnetization components have been determined by using the graphical Zijderveld method (Zijderveld, 1967) and the difference vector method of Hoffman and Day (1978).

Tectonic correction (= bedding correction taking fold axis plunge into account) was performed on sites where the fold axis plunge was $> 10°$. Stepwise unfolding was carried out for three sites where both flanks of the anticline/syncline structure have been sampled (Figs. 4 and 6).

Measurements of anisotropy of magnetic susceptibility (AMS) were made to evaluate the effects of internal rock deformation using a Digico anisotropy meter and a KLY2 Kappabridge, standardized after Hrouda et al. (1983). The following AMS parameters were used:

degree of anisotropy	$P = k_{max}/k_{min}$
magnetic foliation	$F = k_{int}/k_{min}$
magnetic lineation	$L = k_{max}/k_{int}$
q-factor	$q = (k_{max} - k_{int})/[(k_{max} + k_{int})/2 - k_{min}]$

($q > 1$ = prolate susceptibility ellipsoid, $q < 1$ = oblate susceptibility ellipsoid)

For the IRM measurements a Brucker magnet with a maximum field strength of 1.2 T and a

TABLE 1

Paleomagnetic results for Devono-Carboniferous sediments from the Ardenno-Rhenish Massif

Site	Sample		ChRM-directions [a]								Pole positions [a]								Foldtest	
	n	N	Before BTC				After BTC				Before BTC				After BTC				F	F^*
			Dec (°)	Inc (°)	k	α_{95} (°)	Dec (°)	Inc (°)	k	α_{95} (°)	Long°E	Lat°S	dp(°)	dm (°)	Long°E	Lat°S	dp(°)	dm (°)		
Lower Carboniferous																				
WAL	39/85	35/78	192.2	−0.6	32	4.5	195.1	−8.4	19	5.1	350.4	38.6	2.2	4.5	345.6	41.9	3.0	5.9	0.59	1.76
BIN	15/32	09/20	191.0	3.0	36	8.7	196.0	−17.0	26	10.4	352.4	36.9	4.4	8.7	343.2	45.9	5.6	10.8	0.72	3.37
CAP	09/18	09/18	232.0	31.0	51	7.2	219.4	−8.1	44	7.9	315.8	8.8	4.5	3.1	317.0	33.2	4.0	7.9	0.36	3.37
ROY	13/23	13/22	214.8	25.1	349	2.2	207.7	−0.7	333	2.3	329.1	19.5	1.3	2.4	333.0	34.7	1.2	2.3	0.95	2.66
ANS	35/54	32/49	231.4	4.6	53	3.6	235.2	1.6	11	8.3	307.8	21.6	1.8	3.6	303.5	20.8	4.2	8.3	0.21	1.33
ONO	11/17	09/15	210.8	8.3	38	9.0	211.3	−10.2	42	8.1	328.8	29.3	4.6	9.1	323.9	37.6	4.1	8.2	1.11	3.37
LFO	09/15	08/13	216.5	−4.8	40	8.8	218.7	−2.5	52	7.7	320.7	33.3	4.4	8.8	319.0	31.2	3.9	7.7	1.30	3.70
CL		7	212.3	9.8	15	16.1	212.0	−6.6	28	11.6	328.4	28.0	8.2	16.2	324.8	35.7	5.9	11.7		
Upper Devonian																				
MOR	16/25	12/18	212.1	33.3	3	32.5	234.8	38.7	4	26.1	334.4	15.4	21.0	36.9	316.5	2.9	18.4	31.0	1.33	2.79
MON	15/25	11/19	191.1	1.2	11	14.4	201.7	29.8	7	18.5	351.4	38.0	7.2	14.4	343.2	20.8	11.4	20.5	0.64	2.94
GER	14/24	11/16	214.1	−0.6	16	11.7	216.2	−8.6	12	13.9	324.2	32.1	5.9	11.7	319.8	34.8	7.1	14.0	0.75	2.94
DU		3	202.6	0.3	25	52.5	209.4	10.7	8	106	337.3	35.8	26.3	52.5	332.3	28.3	54.5	107		
Middle Devonian																				
VER	12/27	12/25	198.4	19.2	173	3.3	195.7	−20.4	118	4.0	345.7	27.2	1.8	3.4	342.8	47.4	2.2	4.3	0.68	2.79
BAR	20/32	20/31	212.9	25.3	288	1.9	204.9	−5.8	313	1.9	328.6	20.4	1.1	2.1	330.5	38.4	0.9	1.8	1.09	2.15
TIL	31/43	25/39	209.2	5.9	15	7.8	212.2	−0.4	6	12.8	331.0	30.9	4.0	7.9	326.3	32.7	6.4	12.8	0.40	2.04
HEY	33/52	24/44	197.2	2.1	25	5.7	204.1	−12.6	16	7.1	344.0	36.5	2.8	5.7	332.7	41.6	3.7	7.3	0.64	1.37
DUR	20/34	18/28	215.7	−3.1	15	9.4	228.1	7.9	4	20.4	321.7	32.6	4.7	9.4	312.3	21.8	10.3	20.5	0.27	2.26
TAF	19/31	18/29	233.6	22.9	76	4.0	230.6	−12.7	73	4.1	311.2	12.2	2.3	4.2	303.3	29.3	2.1	4.2	0.96	2.25
MAZ	14/20	14/20	203.0	2.5	63	5.1	203.3	−11.8	44	6.1	336.3	34.6	2.6	5.1	333.0	41.4	3.2	6.2	0.70	2.55
DM		7	209.7	10.9	24	12.6	211.3	−8.2	26	12.1	331.2	28.4	6.5	12.8	325.1	37.6	6.2	12.2		
Lower Devonian																				
TAN	20/26	20/26	219.5	28.0	71	3.9	216.0	−13.1	70	3.9	325.1	16.1	2.3	4.3	318.0	37.0	2.0	4.0	0.99	2.15
PRO	12/19	10/13	221.6	15.6	84	5.3	223.6	−23.3	62	6.2	319.9	21.5	2.8	5.4	306.2	37.9	3.5	6.6	0.74	3.13
REG	15/27	14/23	211.9	27.6	76	4.6	208.5	−8.1	35	6.9	332.5	19.1	2.7	5.0	328.2	37.8	3.5	6.9	0.46	2.55
MER	24/39	23/34	203.4	−8.7	50	4.3	205.6	13.4	10	10.3	334.6	39.8	2.2	4.3	336.4	28.5	5.4	10.5	0.20	2.10
(EMB	09/18	09/13	217.6	−7.0	4	29.7	211.2	6.0	16	13.5	319.4	33.6	15.0	29.9	328.9	30.2	6.8	13.6	4.00	3.37)
DL		4	214.1	15.6	18	22.0	213.4	−7.8	22	20.0	327.7	24.2	11.6	22.6	322.6	35.8	10.2	20.2		

[a] BTC = bedding/tectonic correction.

CL = Lower Carboniferous; DU = Upper Devonian (Famennian); DM = Middle Devonian (Givetian–Frasnian); DL = Lower Devonian (Emsian–Eifelian). n, N = number of samples/specimen analyzed and used to compute the site mean; Dec, Inc = declination, inclination of site mean direction (in situ and after bedding/tectonic correction); k, α_{95} = precision parameter, radius of the 95%-confidence circle; Long°E, Lat°S = longitude, latitude of the Virtual Geomagnetic Pole; dp, dm = semiaxis of the 95%-oval of confidence; F = fold test quotient $k_{Corr}/k_{in\ situ}$; F^* = fold test quotient for 99%positive McElhinny foldtest.

Molspin impulse magnetizer with a maximum field strength of 0.3 T were used. Thermomagnetic and X-ray diffraction measurements were undertaken with ferrimagnetic enriched samples (magnetic extracts). However, due to their low susceptibilities, most of the measurements were not meaningful. Optical and electron microscopic investigations (Jeol 35CF electron microscope including EDX analyses) were carried out for carbonates and red sandstones. All details of experimental procedures and data analyses are summarized in Thominski (1991).

Magnetic and magnetomineralogic results

The paleomagnetic results for the 22 sites from the Ardenno-Rhenish Massif which have been sampled and analyzed systematically are given in Table 1 and Figure 2. From seven Lower Carboniferous Kohlenkalk sites a mean direction of Dec 212.3, Inc 9.8 (in situ) and Dec 212.0, Inc −6.6 (after bedding/tectonic correction) has been calculated. From seven Middle–Upper Devonian (Givetian–Frasnian) carbonate sites a mean direction of Dec 209.7, Inc 10.9 (in situ) and Dec 211.3, Inc −8.2 (after bedding/tectonic correction) has been calculated. From 4 Lower Devonian (Emsian–Eifelian) red bed sites a mean direction of Dec 214.1, Inc 15.6 (in situ) and Dec 213.4, Inc −7.8 (after bedding/tectonic correction) has been calculated (site EMB has not been used, see statistics Table 1). A mean direction from the three Upper Devonian (Famennian) Condroz sandstone sites has been calculated, but it is not well defined (Table 1; Fig. 2).

The Lower Carboniferous carbonates ("Kohlenkalk")

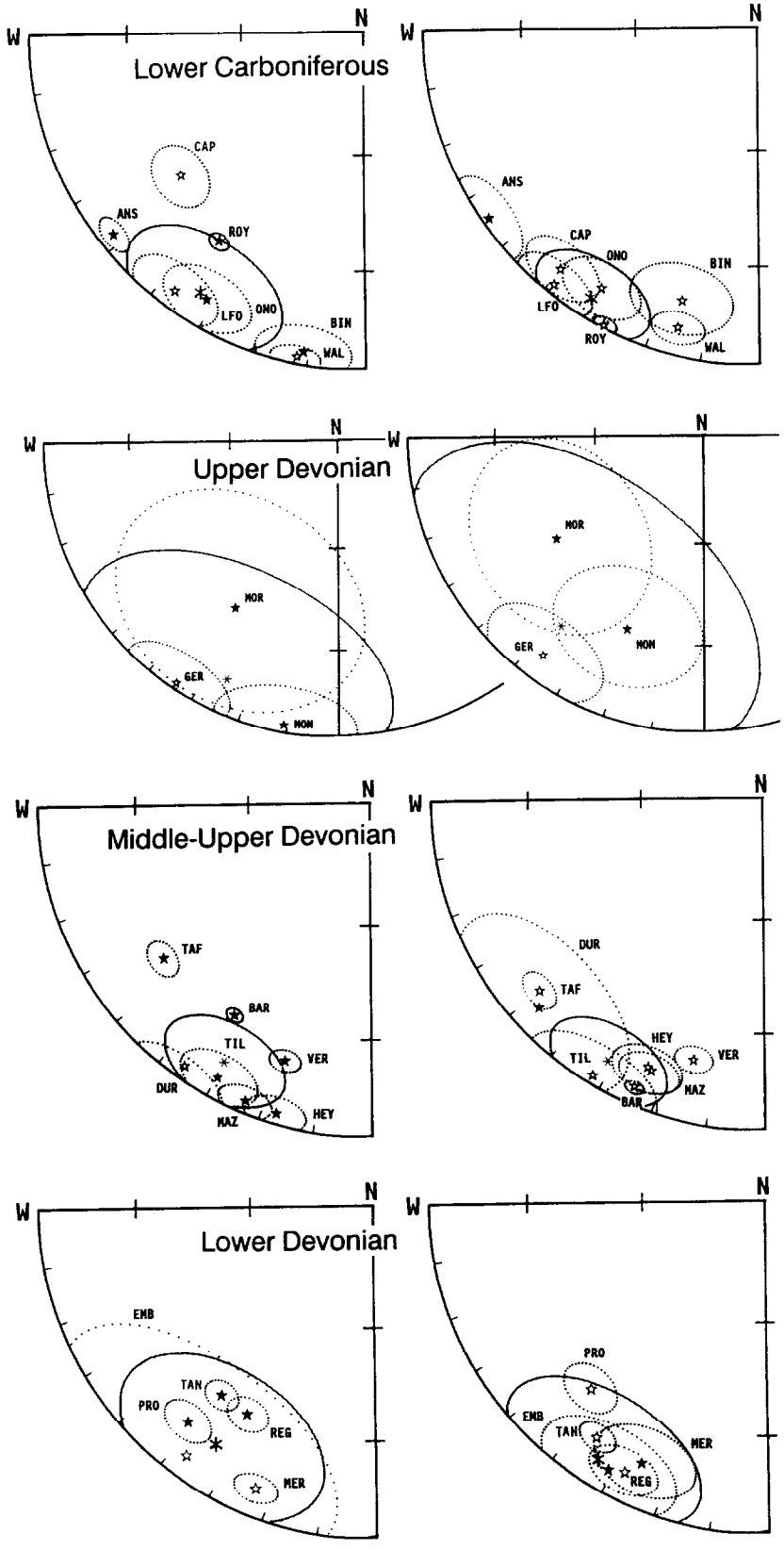
The Lower Carboniferous limestones and dolomites ("Kohlenkalk" facies) carry a two-component magnetization. The stable component is directed shallow to the southwest (Fig. 2) and carried by fine-grained magnetite, as shown by coercivities (20–100 mT; Levi and Merrill, 1978; Bailey and Dunlop, 1983), unblocking temperatures (310–500°C, up to 550°C) and IRM acquisition results (Fig. 3). Cisowski tests (Cisowski, 1981) identify PSD/SD magnetite as the carrier of remanence (Fig. 3d). Similar Cisowski test results have been found for thermoviscous remagnetized Paleozoic limestones of the North American Craton (Onondaga limestone, Trenton limestone, see Jackson, 1990).

Pyrrhotite, which has been shown by Salmon et al. (1988) and Addison et al. (1985) to be the carrier of Permo-Carboniferous remagnetization in similar anoxic Devonian and Carboniferous carbonates, can be excluded as a remanence carrier in the Kohlenkalk. This was determined by the magnetic and optical results and by EDX analyses. Magnetite in the microscopic grain size range occurs as subangular elongated (up to 10×50 μm) and isometric (up to 50 μm) grains and are of detrital origin. No indication of a diagenetic magnetite phase (McCabe et al., 1983; Bachtadse et al., 1987; Salmon et al., 1988; Jackson et al., 1988) was found in the Lower Carboniferous polished sections.

Negative fold tests indicate that the Kohlenkalk magnetization is syn- to post-tectonic (Table 1; Fig. 4). The Lower Carboniferous sediments acquired a thermoviscous remanence [TVRM—Chamalaun (1964), Pullaiah et al. (1975) and Kent (1985)] during the course of their burial in the Variscan orogeny. Assuming burial temperatures of 100–200°C [vitrinite reflectance $R_{max} = 2.8$ for the Inde Syncline and $R_{max} = 3.6$ for the Dinant Syncline—Teichmüller and Teichmüller (1979)] and a burial duration of 10–50 Ma, then a TVRM, carried by magnetite is acquired, which is unblocked at temperatures of 375–450°C (1 hour heating interval) following the calculations of Middleton and Schmidt (1982). These temperatures agree with the unblocking temperatures of the Lower Carboniferous carbonates of the Inde and Dinant synclines investigated here. Lower unblocking temperatures for sites ONO and LFO north of the Aachen–Midi Overthrust corresponds well to the TVRM remagnetization conditions on the stable shelf of the Brabant Massif, as the sediments were buried at shallower depths.

The Middle and Upper Devonian (Givetian–Frasnian) carbonates

The Middle and Upper Devonian (Givetian–Frasnian) reefal limestones and dolomites show


Lower Carboniferous
W
N
CAP
ANS
ROY
LFO
ONO
BIN
WAL
Upper Devonian
MOR
GER
MON
Middle-Upper Devonian
TAF
BAR
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VER
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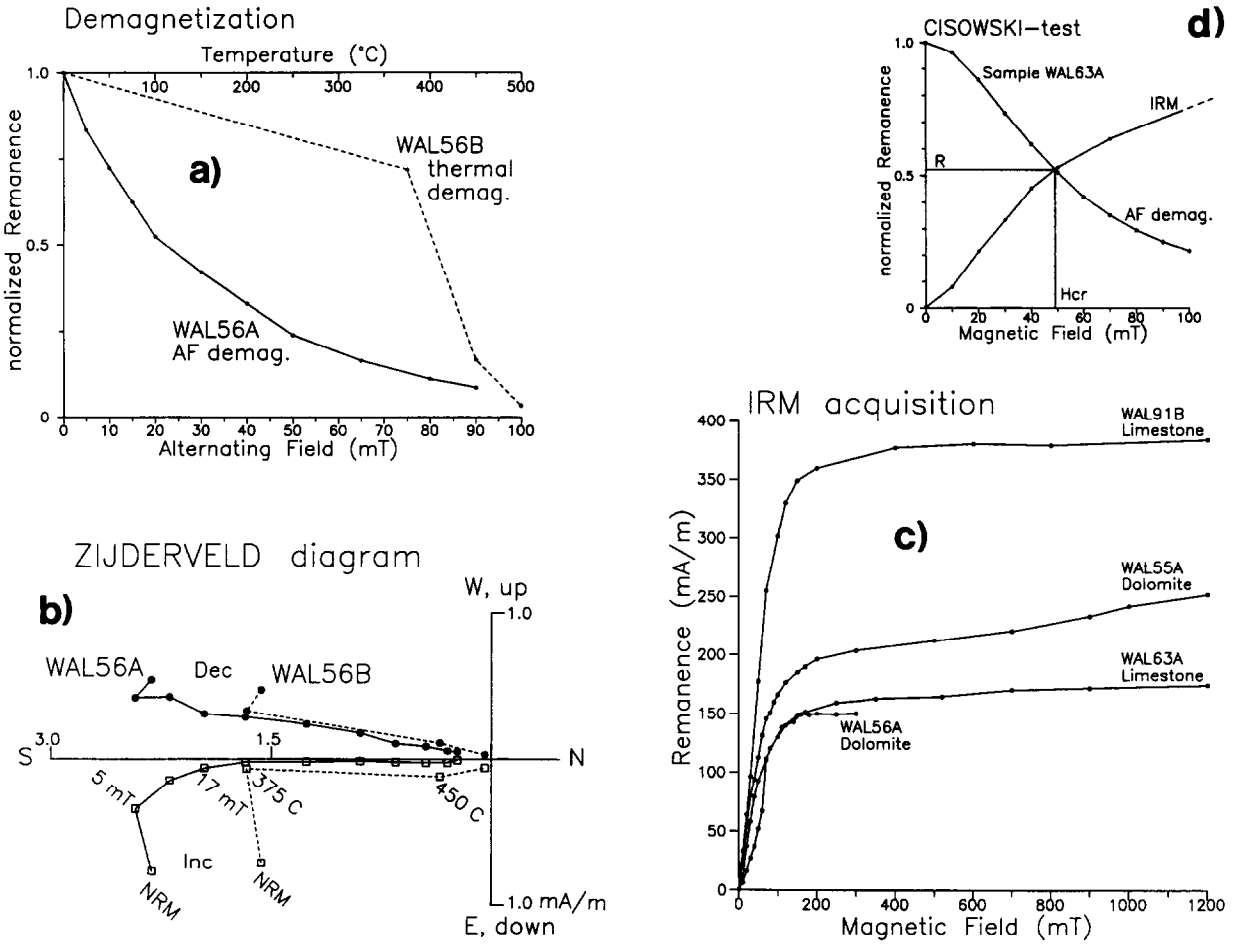


Fig. 3. Magnetic results for Lower Carboniferous carbonates of site WAL (Kohlenkalk). (a, b) Alternating field and thermal demagnetization identify a stable remanence direction with coercivities of 20–90 mT and unblocking temperatures of 375–500°C. (a) Demagnetization curves. (b) Zijderveld orthogonal projection, in situ coordinates. (c) IRM acquisition curves from limestones and dolomites are typical for magnetite; samples which are not saturated in fields < 350 mT contain goethite. (d) Cisowski test of sample WAL63A shows that the magnetite is in the SD-PSD grain size range.

high NRM intensities of around 3 mA/m, which are independent of facies (reef biostromes, laminated lagoonal and terrestrial influenced carbonates). The Devonian carbonates are of the same magnetomineralogy as the Lower Carboniferous carbonates: pyrite and goethite are the dominating ore minerals in all polished sections; pyrrhotite is absent, hematite is rare and magnetite can be found in the form of detritic enrichments. Similar to the Kohlenkalk, coercivities and unblocking temperatures, IRM-acquisition curves, Q-factors and Cisowski-tests identify fine-grained (SD/PSD-) magnetite as the carrier of the shallow SW-directed, stable magnetic directions (Figs. 2 and 5). Fold tests verify that the Givetian–Frasnian magnetization is syn- to post-tectonic;

Fig. 2. Site mean directions of characteristic remanent magnetization (equal area projection, lower hemisphere; closed/ open symbols represent positive/negative inclination, respectively) and resulting formation mean directions (*) for the Lower Carboniferous Kohlenkalk carbonates, the Upper Devonian (Famennian) Crondoz sandstone, the Middle–Upper Devonian (Givetian–Frasnian) carbonates and the Lower Devonian (Emsian–Eifelian) red beds (left = in situ; right = after bedding/tectonic correction). See also Table 1.

for site Tilff (TIL) the fold test becomes positive at 30–40% unfolding (Fig. 6a), for site Durbuy (DUR) direction scatter becomes a minimum at 20–30% unfolding (Fig. 6b).

A strong indication for the thermoviscous remagnetization of the Carboniferous and Devonian carbonates are the nearly identical formation directions and statistical parameters (Table 1). Higher unblocking temperatures and coercivities of the Devonian carbonates correspond to a thermoviscous remagnetization, resulting from deeper and longer burial compared with the Kohlenkalk. For site MAZ, situated on the shelf of the Brabant Massif, temperatures of 100–200°C at a burial time between 50 and 10 Ma are sufficient to produce the TVRM remagnetization.

The Upper Devonian (Famennian) Condroz sandstone

The Upper Devonian graywackes and shales are magnetically weak and difficult to analyze (mean NRM = 0.88 mA/m). Only three sites of Condroz sandstone yielded consistent results (MOR, MON, GER; Table 1; Fig. 2). IRM acquisition measurements as well as coercivities and

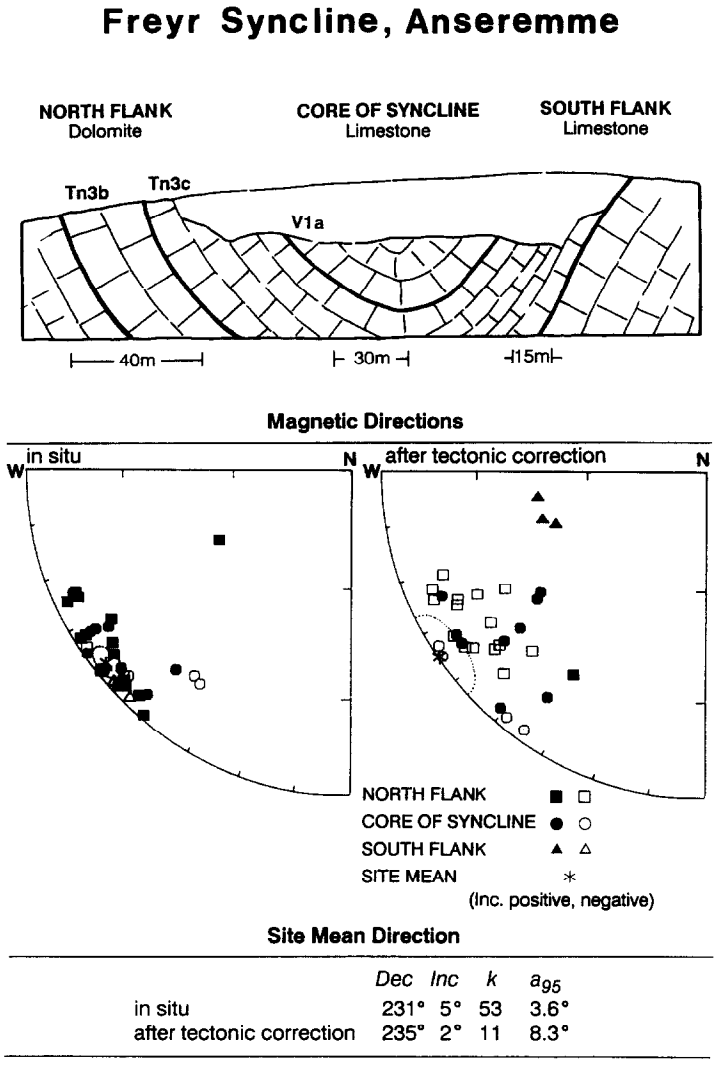


Fig. 4. Negative fold test result for Lower Carboniferous site ANS: stepwise unfolding of the Freyr Syncline shows that direction scatter becomes a maximum at 100% unfolding. Above: geological profile of the Freyr Syncline with sampling intervals. Below: equal area plot (lower hemisphere) of stable magnetic directions, in situ and after tectonic correction (= 100% unfolding).

unblocking temperatures identify magnetite (MOR) and also magnetite/hematite (MON, GER) as carriers of stable remanence directions (Fig. 7). Due to the agreement of the Condroz site mean directions with the remagnetization direction of the carbonates (Table 1), a similar TVRM acquisition can be assumed.

With a few exceptions, in all the formations investigated the magnetizations are of inverse polarity, corresponding to the Kiaman paleofield configuration in the Permo-Carboniferous. Only in the Famennian Condroz sandstone do some antiparallel and intermediate remanence directions occur which are possibly recent chemorema-

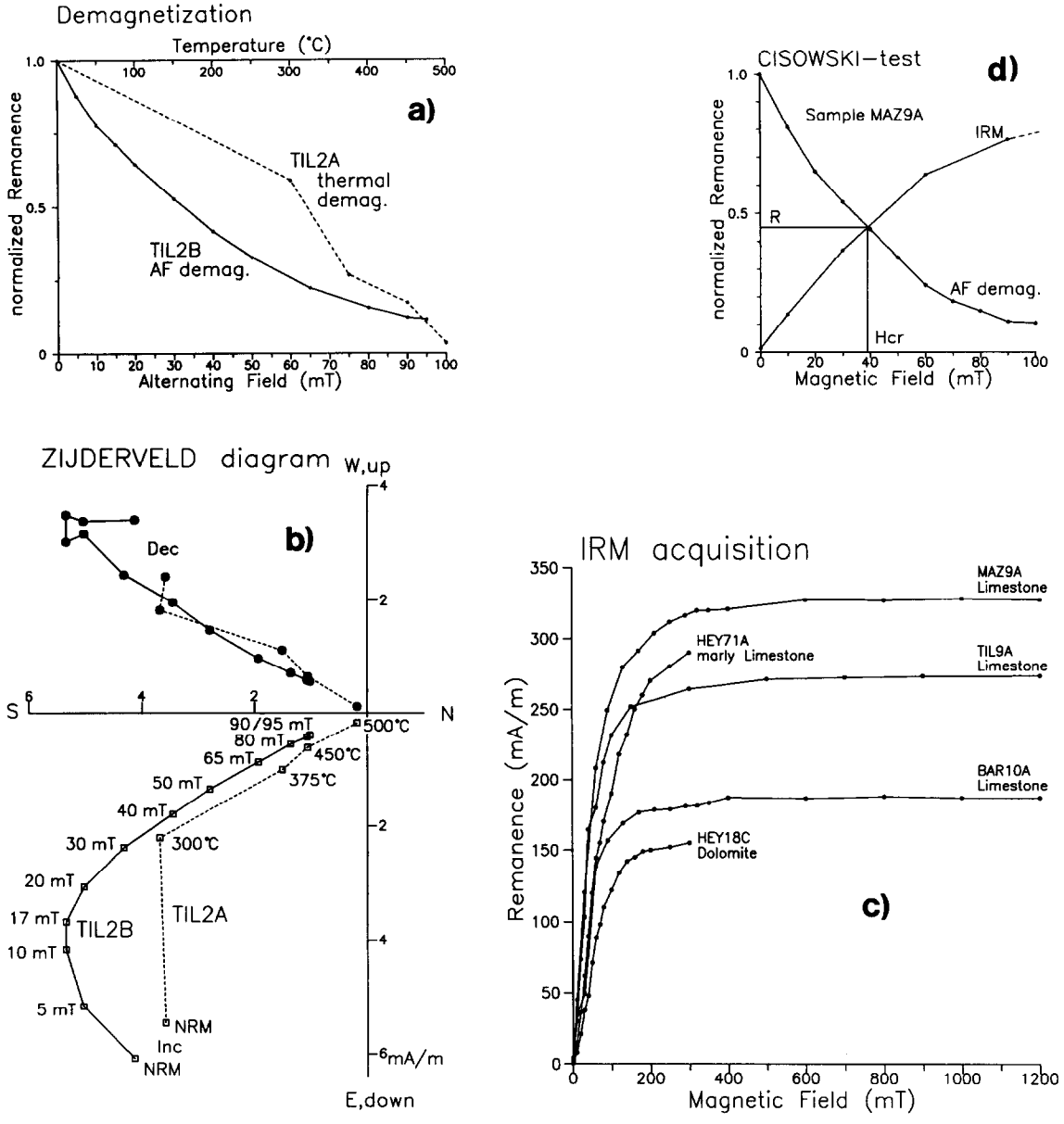
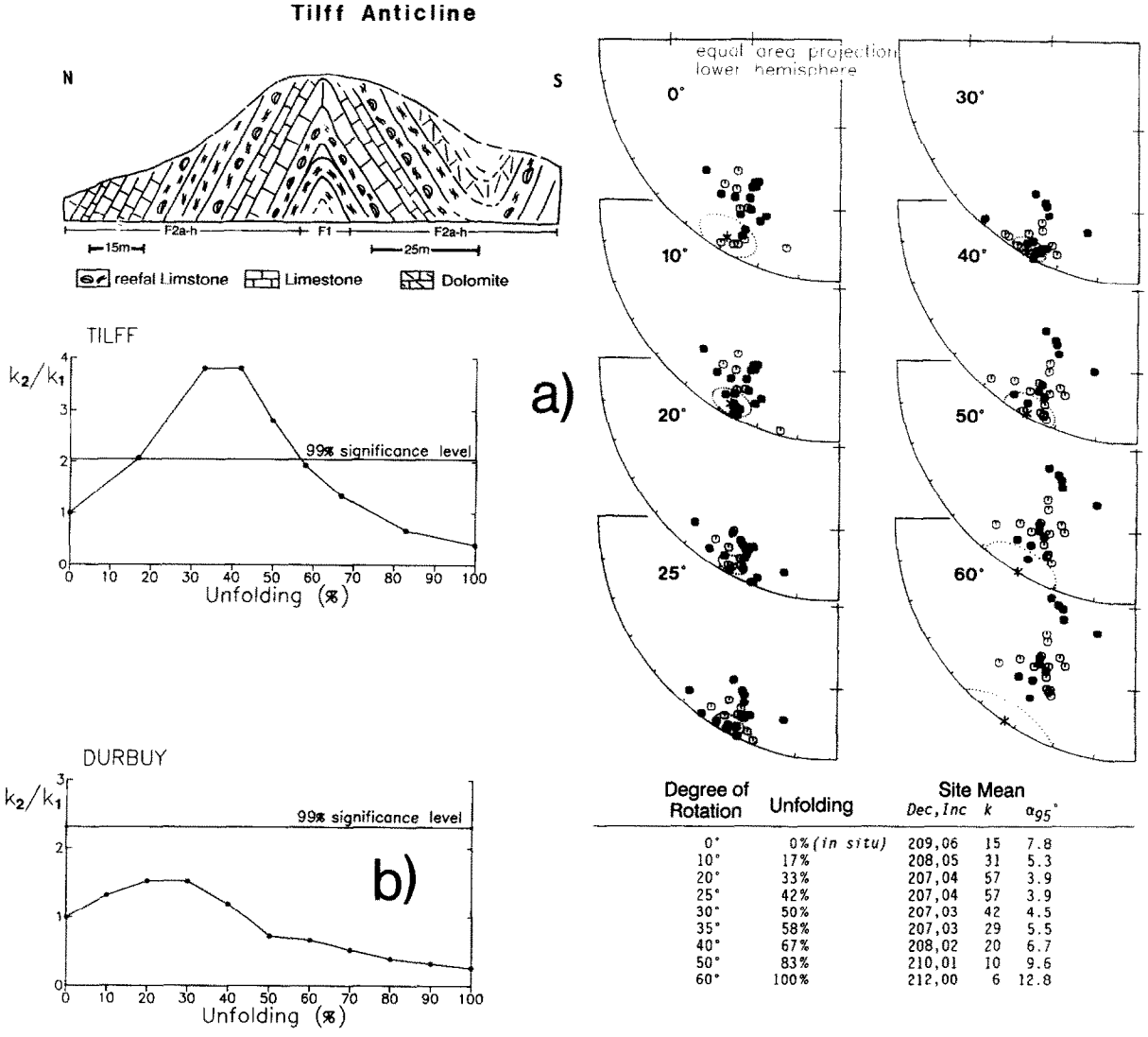


Fig. 5. Magnetic results for Givetian–Frasnian carbonates. Alternating field and thermal demagnetization identify a stable remanence direction with coercivities of 20–95 mT and unblocking temperatures of 300–500°C typical for magnetite (a, b; in situ coordinates). Identification of magnetite is confirmed by IRM acquisition measurements (c) and Cisowski test results (d).

nent magnetizations carried by hematite. The detailed magnetostratigraphy of the Devono-Carboniferous boundary proposed by Kolesov (1984) is not confirmed by the results of this study.

The Lower Devonian (Emsian–Eifelian) red beds

The Lower Devonian Emsian–Eifelian red sandstones carry a two-component magnetization. SW–SSW-directed stable remanence directions (Fig. 2) are isolated between 450 and 680°C and resided in hematite pigment, as shown by the continuous, only slightly varying unblocking temperatures (Irving and Opdyke, 1965; Collinson, 1974; Fig. 8). The square-shouldered unblocking spectra over a small temperature range of numerous samples indicate that the hematite is relatively uniform in composition and grain size. IRM acquisition measurements show that the hematite occurs predominately in the grain size fraction <



Degree of Rotation	Unfolding	Site Mean Dec, Inc	k	α_{95}°
0°	0% (in situ)	209,06	15	7.8
10°	17%	208,05	31	5.3
20°	33%	207,04	57	3.9
25°	42%	207,04	57	3.9
30°	50%	207,03	42	4.5
35°	58%	207,03	29	5.5
40°	67%	208,02	20	6.7
50°	83%	210,01	10	9.6
60°	100%	212,00	6	12.8

Fig. 6. Fold test results for Givetian–Frasnian sites TIL and DUR (Ourthe Valley, Dinant Synclinorium). (a) Stepwise unfolding of the Tilff Anticline (geological profile with sampling intervals) and resulting direction scatter of stable remanence directions. At 30–40% unfolding the fold test (McElhinny, 1964) becomes positive at the 99%-significance level. (b) Stepwise unfolding of the Durbuy Anticline shows that the 99%-significance level for positive fold test is not reached, but direction scatter becomes a minimum at 20–30% unfolding.

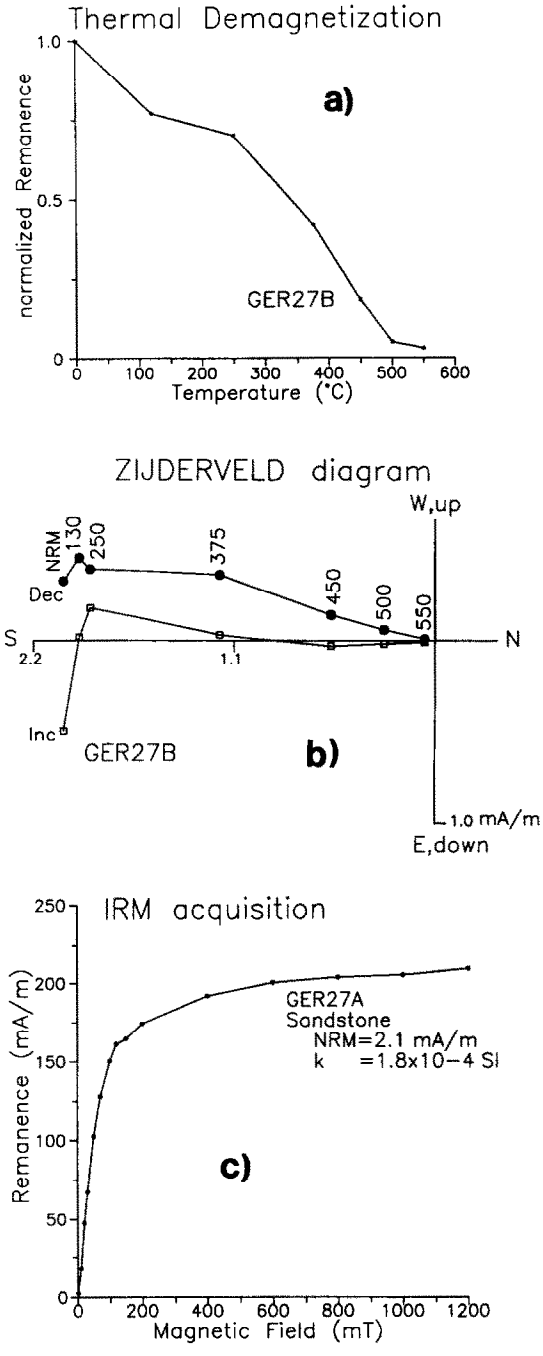


Fig. 7. Magnetic results for Upper Devonian Condroz sandstones of site GER. Thermal demagnetization and IRM acquisition identify magnetite as the carrier of a shallow, SSW-directed stable remanence direction.

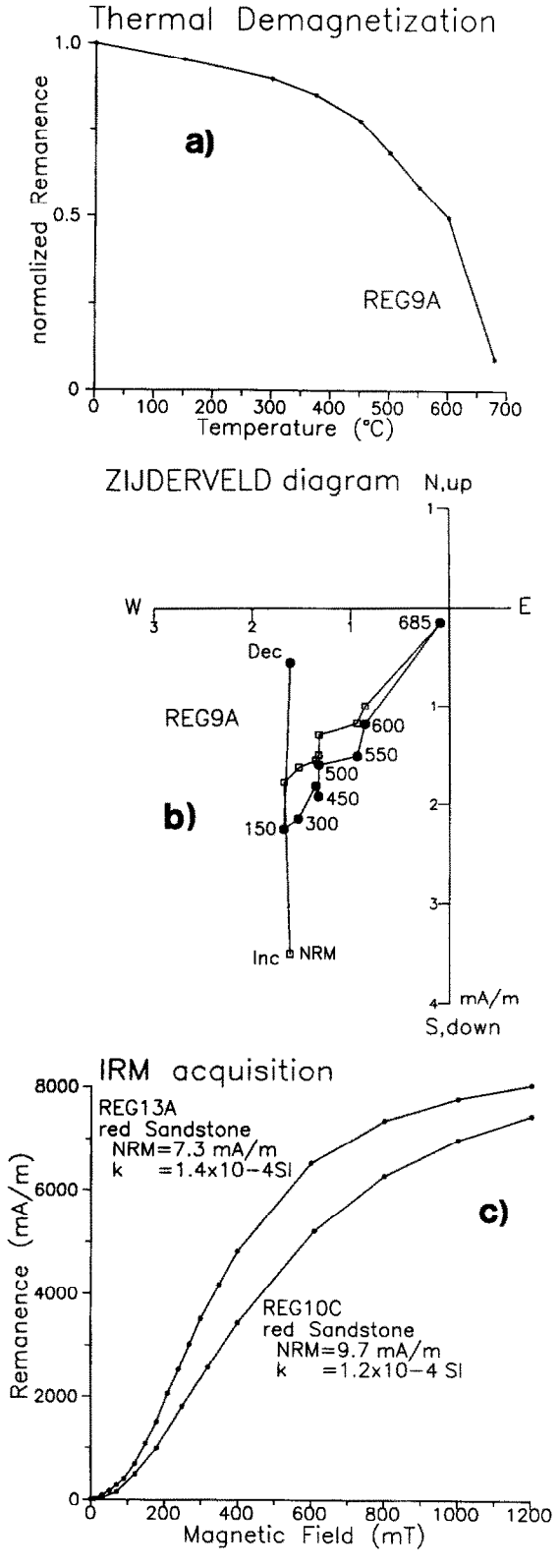


Fig. 8. Magnetic results for Lower Devonian red sandstones of site REG. (a, b) Thermal demagnetization identify a stable remanence direction with unblocking temperatures typical for hematite pigment (in situ coordinates) (Collinson, 1974). Hematite pigment in the 1–20 μm grain size range (Dunlop, 1972; Dankers, 1988) is confirmed by IRM acquisition results (c).

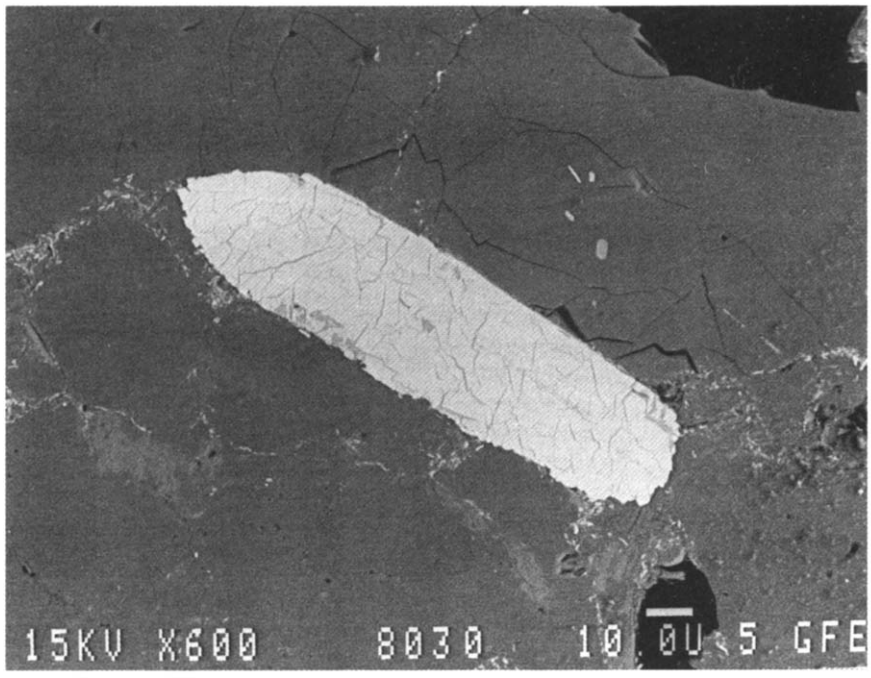


Fig. 9. Hematite is dominating in the red sandstone polished sections (specularite < 50 μm and pigment < 5 μm). Magnetite is rare and can be found in form of oxidized detritic grains with exsolution laminae.

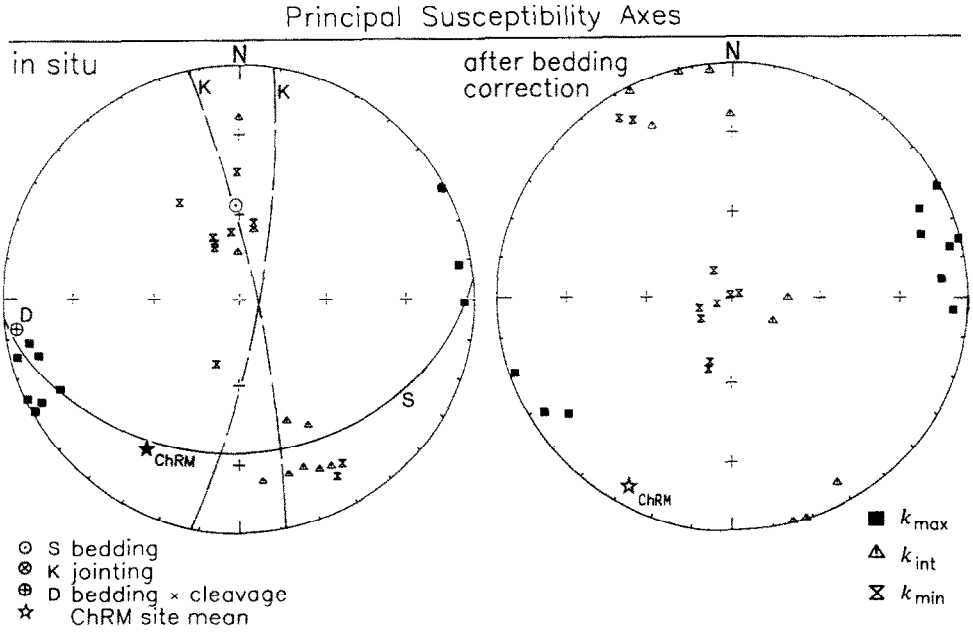


Fig. 10. Equal area projection of principal susceptibility axes of red sandstones from site REG. The k_{min}-axes are normal, the k_{int}- and k_{max}-axes are parallel to the bedding plane (= sedimentary-diagenetic fabric). The k_{max}-axes are parallel to the fold axis which is indicated by the bedding × cleavage line (= tectonic fabric).

TABLE 2

Anisotropy of magnetic susceptibility parameters for red bed sites

Site	Suscep. 10^{-5} SI	Anisotropy factor P	Foliation F	Lineation L	q-factor
Regissa (REG)	9.39	1.028	1.017	1.011	0.61
Mery (MER)	18.12	1.023	1.014	1.009	0.60
Embourg (EMB)	9.88	1.019	1.011	1.008	0.66
Tailfer (TAN)	17.35	1.040	1.035	1.005	0.15

20 μm (Fig. 8c). Hematite pigment dominates in the polished sections and platy hematite in the range < 1 μm was identified by electron microscopy. Magnetite is found to be more or less oxidized (Fig. 9).

AMS parameters are similar for all red beds of the eastern Dinant Synclinorium (REG, MER, EMB; Table 2). The q-factors are still in the range which is typical for non-deformed sediments (Hamilton and Rees, 1970; Hrouda, 1982). Deviating values however were found for site TAN, which is possibly due to the conglomeratic sandstones. The magnetic fabric of sites REG (Fig. 10) and MER (Fig. 11) shows both pre- and post-tectonic elements: k_{max} and k_{int} orientated in the bedding plane (= sedimentary-diagenetic fabric) and k_{max} parallel to the fold axis (= tectonic fabric). AMS parameters (Table 2) and orientation of principal susceptibility axes identify a sedimentary-diagenetic hematite fabric, which is only slightly tectonically deformed, characteristic for Old Red Sandstones (Fuller, 1963; Collinson, 1965; Cogné, 1988). The tectonic deformation can be classified as Stage 1 to 2 following Kligfield et al. (1983); therefore compensation of internal rock deformation is not thought to be important and can therefore be ignored.

AMS and stable remanence directions are carried by the the same fine-grained hematite phase, which has formed during diagenesis and orientated in the mostly bedding-parallel cleavage plane. A synsedimentary orientation of planar hematite grains can be excluded for the coarse-grained and conglomeratic sandstones. Also a magnetite DRM/PDRM can be excluded, as the original detritic magnetite has been oxidized and is thus unable to yield paleofield data (Heider and Dunlop, 1987). A TVRM can be omitted

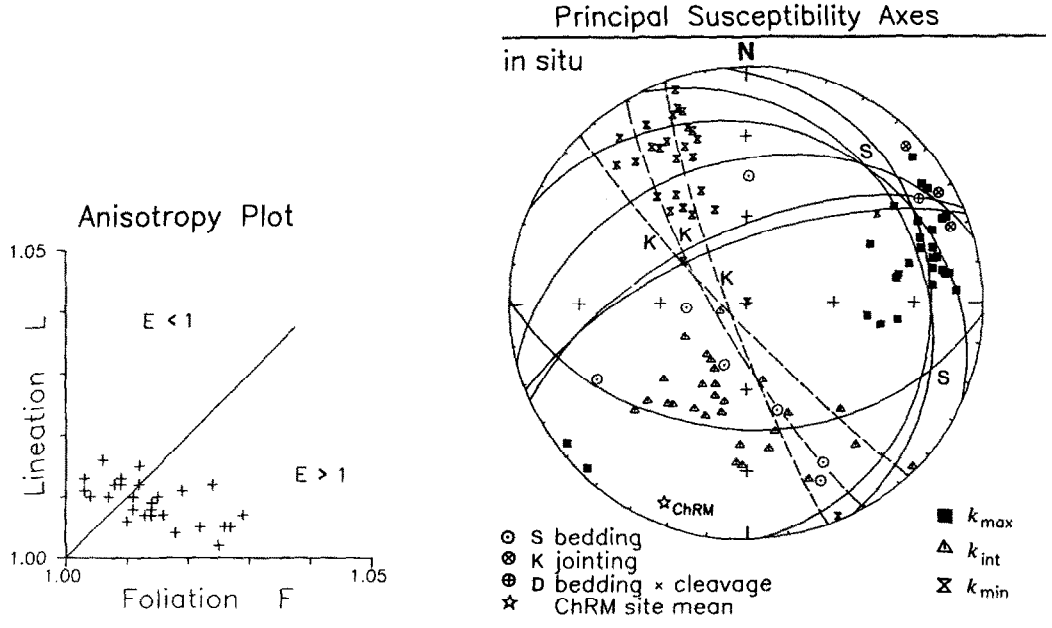


Fig. 11. Equal area projection of principal susceptibility axes and anisotropy plot of red sandstones from site MER. As for site REG, magnetic fabric and orientation of principal susceptibility axes identify a sedimentary-diagenetic magnetic fabric, which is only slightly tectonically deformed (the orientation of k_{int} normal to bedding is not significant because of the weak foliation: $F = 1.014$, $L = 1.009$).

TABLE 3

Rhenohercynian and Saar–Nahe region Permo-Carboniferous pole positions

Region	°E	°S	Age	Ref.	Code
Malmedy conglomerate	346	46	P	Nairn, 1960	1
Winweiler/Grenzlager volcanics	348	48	Pl	Nijenhuis, 1961	2
Saar–Nahe red beds	5	36	Stephanian	Nairn, 1960	3
Saar–Nahe magmatites	356	48	Pl	Roche et al., 1962	4
St. Wendel sandstone, Saar–Nahe	5	45	P	Nairn, 1960	5
Eifel sandstone	343	42	*	Nairn, 1960	6
Harz graywackes and magmatites	332	32	(*?)	Rother, 1971	7
Harz graywackes and limestones	354	46	*	Bachtadse, 1984	8
Sauerland graywackes and volcanics	3	47	*	Bachtadse, 1984	9
Deckdiabase, Lahn Dill	348	41	*	Doutch et al., 1985	10
Kohlenkalk, Inde syncline	2	48	* (Cu-Pl)	Nowaczyk, 1985	11
Frasnian limestone, Inde syncline	345	44	(*?)	Nowaczyk, 1985	12
Lower Devonian graywackes, Inde syncline	1	45	* (Cu-Pl)	Nowaczyk, 1985	13
Limestones and sandstones, Ardennes	323	38	* (Cu-Pl)	Edel and Coulon, 1987	14
Magmatites, Brabant Massif	333	39	* (Cu-Pl)	Edel and Coulon, 1987	15

*: Permo-Carboniferous remagnetization.

because of the high unblocking temperatures, corresponding to remagnetization temperatures of 380–680°C (Pullaiah et al., 1975; Kent and Miller, 1987), which were not attained in the Variscan foredeep. The Emsian–Eifelian red sandstones carry a chemoremanent magnetization, which is typical for red beds (Kent and Opdyke, 1985; Kent and Miller, 1987). The resulting formation direction differs only slightly from the remagnetization direction of the carbonates (Table 1). Remagnetization in the Permo-Carboniferous Kiaman paleofield is confirmed by the inverse polarities found exclusively in the Emsian–Eifelian red beds.

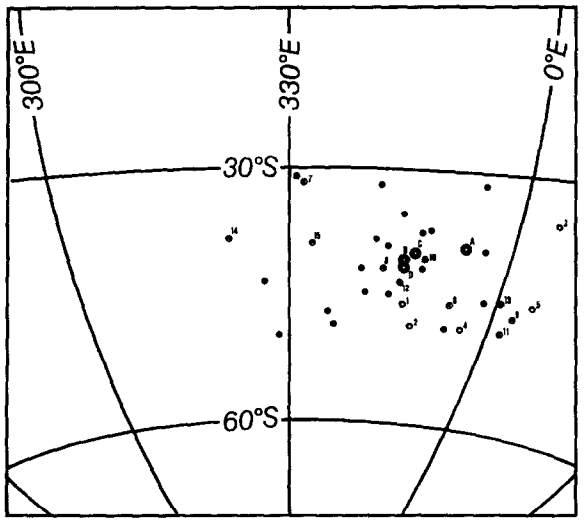


Fig. 12. European Permo-Carboniferous pole positions. Unnumbered circles: poles of Permo-Carboniferous rocks from the Bohemian Massif, Central Massif, Black Forest, Southern France, Southern England and Southern Sweden. Numbered circles: poles of Permo-Carboniferous rocks (poles 1–5, Table 3) and of Paleozoic rocks which have been remagnetized in the Permo-Carboniferous (poles 6–15) from the Rhenohercynian and the Saar Nahe region. *A*, *B*, *C* and *D*: Permo-Carboniferous pole compilations (Table 4).

The lower Paleozoic sediments (Cambrian–Silurian)

Lower Paleozoic sedimentary rocks were sampled at eighteen sites in the Brabant Massif, the Stavelot–Venn Massif and the Rocroi Massif (Fig. 1). The quartzitic sandstones and shales proved to be unsuitable for paleomagnetic investigations due to their generally low NRM intensities and inconsistent demagnetization results.

Paleomagnetic interpretation of the virtual geomagnetic poles

European apparent polar wander paths (Briden et al., 1973; McElhinny, 1973; Duff, 1980) are based mainly on data, for which primary magneti-

TABLE 4

Permo-Carboniferous pole compilations

Region	°E	°S	Age	Ref.	Code
"Stable Europe"	354	39	Stephanian-Pl	Duff, 1980	A
Europe	346	41	Pl	Tarling, 1983	B
North European Platform	347	40	Pl	Khramov, 1987	C
Europe	346	42	Cu-Pl	This study (Fig. 12)	D

zation is not verified by fold tests or other stability tests. According to the analysis of Briden and Duff (1981) there only remains one key pole for Paleozoic Europe. When referring to new paleomagnetic investigations from North America, which have revised a large number of Paleozoic poles as remagnetizations (Irving and Strong, 1984; Miller and Kent, 1986; Van der Voo, 1988), one must critically assess any comparisions made with a so-called "European apparent polar wander path".

A compilation of Permo-Carboniferous pole positions for Europe is given in Tables 3 and 4 and in Figure 12. The unnumbered poles of Figure 12 (full circles) are based on primary (syngenetic) magnetizations of Upper Carboniferous–Lower Permian rocks from the Bohemian Massif, the Central Massif, the Black Forest, Southern France, Southern England and Southern Sweden (for compilation, see Thominski, 1991). For the Rhenohercynian and Saar Nahe region poles of Variscan remagnetized rocks are also given (see

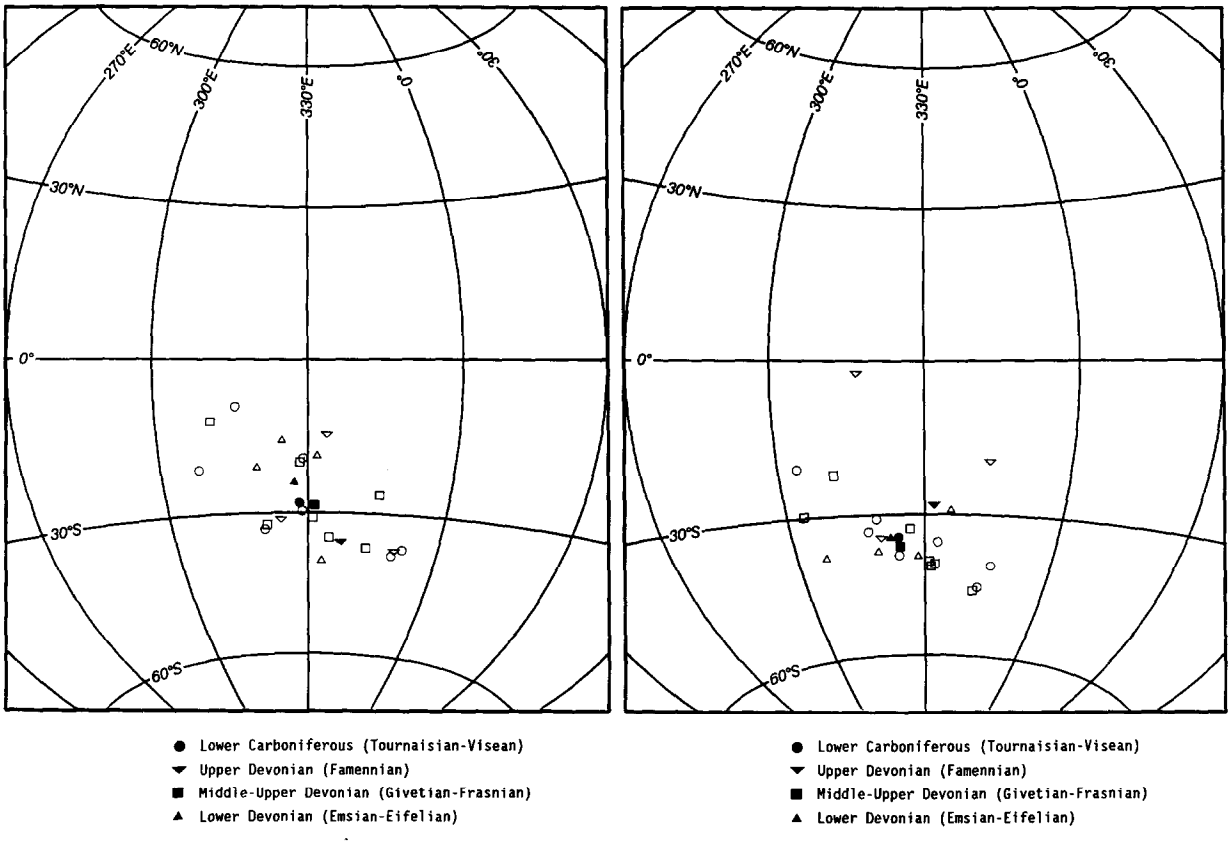


Fig. 13. Virtual geomagnetic poles for Inde Syncline and Dinant–Namur Synclinorium sites (open symbols) and resulting pole positions (full symbols) for the Lower Devonian red beds, the Middle–Upper Devonian carbonates, the Upper Devonian Condroz sandstone and the Lower Carboniferous Kohlenkalk. Left = in situ; right = after bedding/tectonic correction. See also Table 1.

symbols Table 3). The Permo-Carboniferous pole compilations by Duff (1980), Tarling (1983) and Khramov (1987) are marked with the letters A, B and C (Table 4; Fig. 11). Pole D has been calculated from the 24 Late Carboniferous–Early Permian European poles of Figure 12.

The virtual geomagnetic poles (VGPs), determined in this study plot in a region between 310–350°E and 10–40°S (Fig. 13). Bedding correction affects longitude of the poles only slightly but results in a latitudinal shift to more southern positions. Confidence ovals of VGPs of different age (both in situ and bedding corrected, Fig. 2) overlap and there is no grouping of VGPs according to geological age, but to geological structural units (Inde Syncline, Dinant–Namur Synclinorium, see Table 1 and Fig. 13). The Early Carboniferous, Middle–Late Devonian and Early Devonian formation poles differ only slightly. These results clearly indicate the magnetic overprint of the Devono-Carboniferous sediments.

VGPs WAL and BIN of the Inde Syncline and poles HEY and MON of the eastern Dinant Syncline coincide with the Permo-Carboniferous pole of Europe (Table 1; Figs. 12 and 13). The pole of site VER, which is situated near the Burtscheid thrust fault, shows only a slight latidudinal difference; bedding correction shifts the pole to the Permo-Carboniferous reference direction. This can be interpreted as a result of syntectonic magnetization, possibly overprinted by the thrust faulting deformation process. Therefore the rocks from these five sites were completely remagnetized in the Permo-Carboniferous. Similar remagnetization poles have been determined by Nairn (1960) for Lower Devonian sandstones of the Eifel region and by Doutch et al. (1985) for the Lahn–Dill diabase. The Frasnian limestones of the Inde Syncline, which have been investigated by Nowaczyk (1985), are probably also remagnetized (pole 12, Table 3). However, the VGPs of the Dinant–Namur Synclinorium deviate from the Permo-Carboniferous reference direction. These deviations correlate with the bending of geological structures in the Ardenno-Rhenish Massif, as shown by the magnetotectonic analysis.

Magnetotectonic analysis

The magnetic directions determined show a clear correlation with structural trends in the

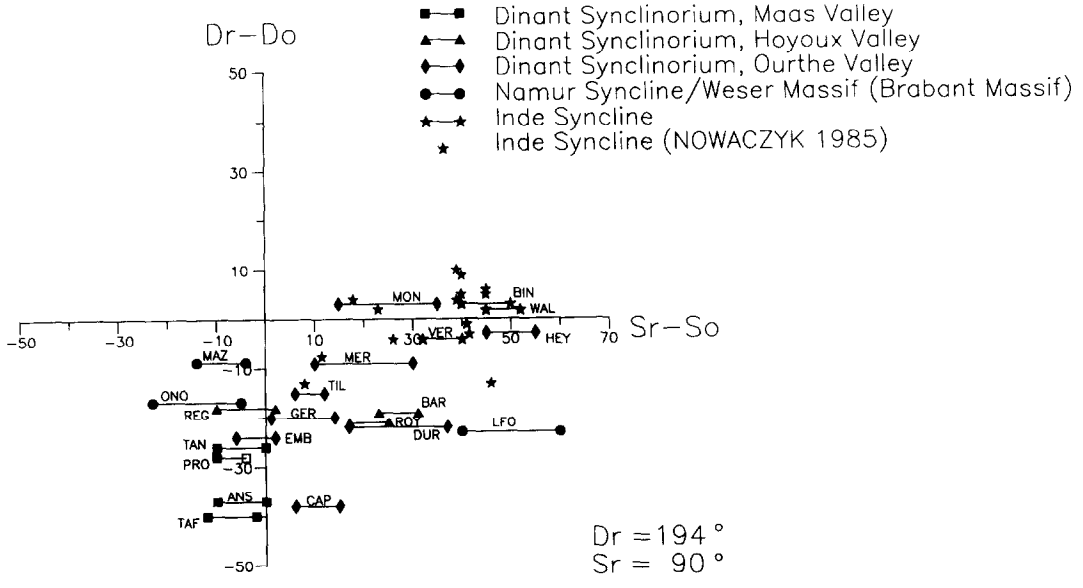


Fig. 14. The declination–strike diagram quantifies the declination and strike variation for individual paleomagnetic sites. D_r = declination reference direction (= 194°, Permian reference direction of Malmedy conglomerate; see Table 3); D_o = mean site declination (in situ); S_r = strike reference direction (= 90°); S_o = mean site strike direction. $\Delta D = D_r - D_o$; $\Delta S = S_r - S_o$; $\Delta DS = S_o - D_o + 180°$.

Ardenno-Rhenish Massif (Fig. 15). From east to west the bending of geological structures is shown by the NE–SW trends of the Linksrheinisches Schiefergebirge (Inde Syncline), the E–W and ENE–WSW trends of the eastern Dinant Synclinorium (Ourthe Valley, Hoyoux Valley) and the ESE–WNW trends of the central Dinant Synclinorium (Maas Valley). These correlations can be quantified in a declination–strike plot (ΔD–ΔS plot, Fig. 14).

The Inde Syncline

The Inde Syncline sites WAL, BIN (Kohlenkalk) und VER (Frasnian) differ only slightly in their declination from the Permo-Carboniferous reference direction of the Malmedy conglomerate ($\Delta D = -4$ to 3°, Figs. 14 and 15; for explanations see Fig. 14). Constant ΔDS values (32 to 36°) and shallow inclinations together with negative fold test results verify the post-tectonic age of the magnetization. Steeper inclinations of site VER are probably caused by an epigenetic dolomitization associated with the Burtscheid thrust fault. The magnetic results for the Inde Syncline sediments determined by Nowaczyk (1985) are similar for the Kohlenkalk (ΔD values about 10°, shallow inclinations of -12 to 2°), whereas the Frasnian carbonates show steeper inclinations (-23 to -12°) and ΔD values of -13 to 4°.

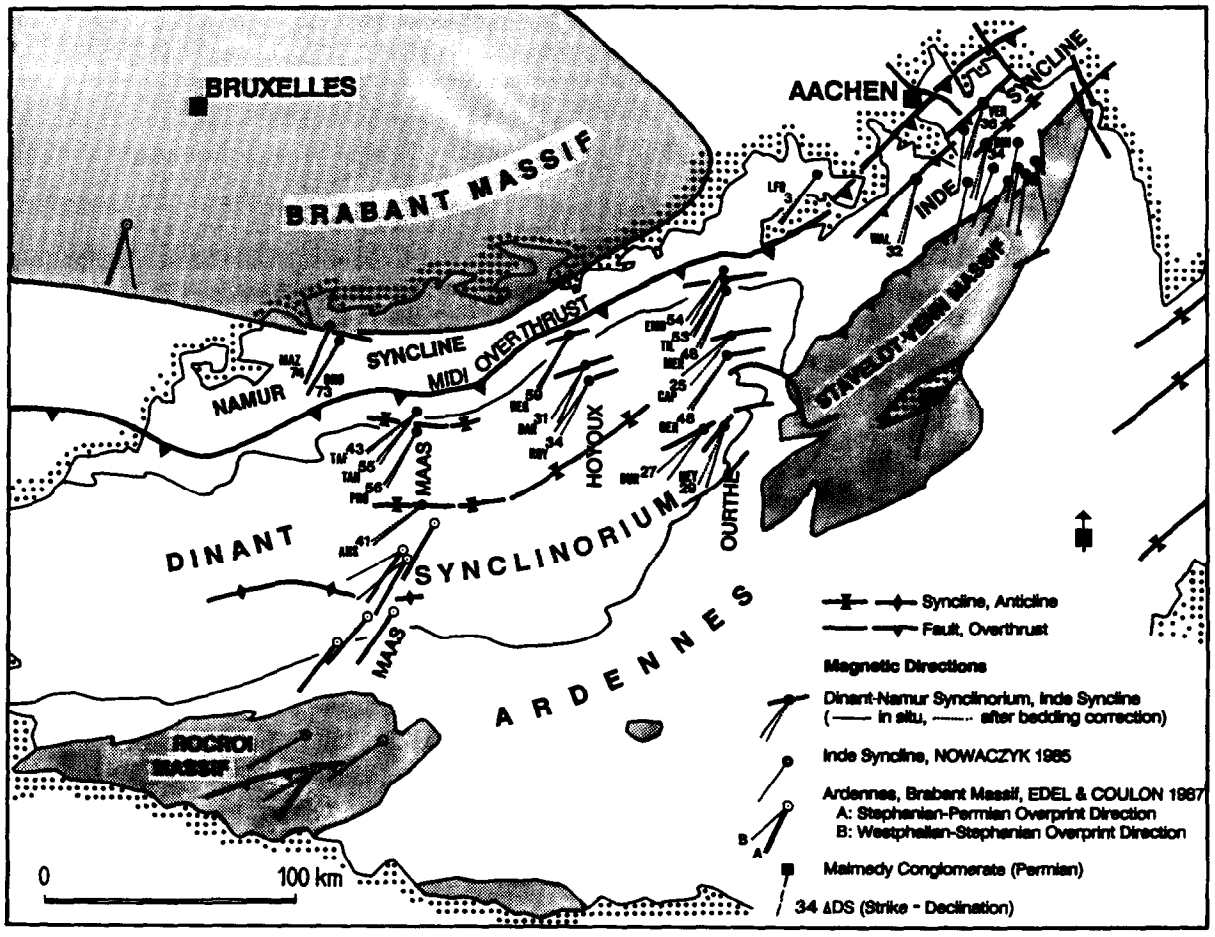


Fig. 15. Magnetotectonic map of the Ardenno-Rhenish Massif showing the relations of structual trends and magnetic declinations [ΔDS = mean site strike direction − mean site declination (in situ) + 180°].

The Dinant Synclinorium

In the eastern Dinant Synclinorium (Ourthe valley profile) two regions can be distinguished magnetotectonically: in the northern part of the Ourthe profile (sites EMB, TIL, MER, GER) geological structures follow an ENE–WSW direction ($\Delta S = 0$–$20°$) with ΔDS values around 50°. In the southern part of the Ourthe profile (sites HEY, DUR) structures follow an NW–SE direction ($\Delta S = 30$–$50°$) with ΔDS values around 28° (Figs. 14 and 15). The ΔS, ΔD and ΔDS values and the shallow inclinations of site HEY (Givetian–Frasnian) agree with those obtained for the Inde Syncline, indicating a Permo-Carboniferous remagnetization. A late tectonic remagnetization, acquired as the strata were folded to 3/4 of the present form of the anticline, was determined for site DUR (Fig. 6). The last phase of the deformation process was accompanied by a dextral shear deformation of about 20°, as shown by the deviation from the magnetic reference direction ($\Delta D = -22°$).

The sites in the northern part of the Ourthe profile (except CAP) show despite their differing lithology (EMB, MER—Lower Devonian red beds; TIL—Frasnian carbonates; GER—Condroz sandstone) the same shallow inclinations (-9 to 6°) and ΔDS values around 50°; their ΔD values vary between -20 and $-10°$. For site TIL a significant late tectonic magnetization was determined (Fig. 6). Assuming that the sediments of the Dinant Synclinorium and Inde Syncline were remagnetized at the same time, then ΔD values of sites EMB, TIL and MER (-25 to $-10°$) indicate a dextral rotation of this region of between 10 and 25°. This probably took place in the last phase of deformation. The deviating values of site CAP (inclination = 31°, $\Delta D = 40°$) can be explained by an early tectonic remagnetization.

The three sites in the Hoyoux valley REG, BAR and ROY show similar magnetic results (inclination = 25–28°; $\Delta D = 18$–$21°$). Steep in situ inclinations and shallow bedding-corrected inclinations point to an early-tectonic remagnetization and a subsequent dextral shear deformation of about 20°.

The central Dinant Syncline (Maas valley profile), however, shows a different picture. Here, the highest ΔD values occur ($\Delta D = -25$ to $-40°$). These deviations from the magnetic reference direction correlate with the trend of geological structures, which shows in relation to the northern Linksrheinische Schiefergebirge a similar dextral rotation. The steeper in situ inclinations of sites TAF, TAN and PRO point to an early tectonic remagnetization as determined for the Hoyoux valley. In contrast, a late tectonic remagnetization is probable for site ANS (inclination = 5°; see also Fig. 4). The greater dextral rotation of this site took place after the Kohlenkalk was folded.

Autochton north of the Midi overthrust

The sites MAZ and ONO, located north of the Faille du Midi, belong to the stable shelf of the Brabant Massif and show the lowest degree of deformation. Based on statistical parameters, the site mean directions are identical (Fig. 2) and agree with the directions determined by Edel and Coulon (1987). The shallow inclinations agree with those from the Inde Syncline and the declination of site MAZ is, on the basis of its statistical parameters, identical with those declinations of the Inde Syncline. These results point to a post-tectonic remagnetization of sites MAZ and ONO. However, it is possible that the ΔD values of these sites are caused by post-Permocarboniferous faulting. The deviating ΔD values of site LFO are probably due to dolomitization and intensive karstification (Poty, 1982).

Conclusions

The paleo- and rock-magnetic investigations have shown that the carbonates are characterized by a thermoviscous remagnetization carried by fine-grained (SD/PSD-) magnetite and that the red sandstones are characterized by a chemoremanent remagnetization carried by hematite pigment. It is assumed, that the TVRM acquisition coincided with the Variscan metamorphism caused by the burial of the sediments. Coalification data (Teichmüller and Teichmüller, 1979) have shown that the temperatures for complete

thermoviscous remagnetization (100–200°C) have been reached. As the folding in the Inde Syncline and in the Dinant–Namur Synclinorium is younger than Westfalian A, the remagnetization must have occurred in late Westfalian–Early Permian times. A synchronous remagnetization of the Rhenohercynian sediments of the Dinant–Namur Synclinorium, and in its eastern continuation the Inde Syncline, also indicates that the metamorphic ages in the Ardenno-Rhenish Massif are varying normal to strike direction but not within strike direction (Ahrend et al., 1983; Pique et al., 1984).

Variscan ages of metamorphism for the Ardennes and the northern border of the Dinant–Namur Synclinorium have been determined at 312 Ma (Michot, 1980) and 297 Ma (Pique et al., 1984), respectively. As the cleavage in the Dinant–Namur Synclinorium is synmetamorphic (Dandois, 1981; Pique et al., 1984), it is probable that the remagnetization of the clastic and carbonate sediments of the Dinant–Namur Basin took place at the same time. This is verified, as neighbouring carbonate and red bed sites (except TAF and TAN) show, within the the accuracy of paleomagnetic measurements, good agreement in their magnetic directions.

The VGPs of the Inde Syncline agree with the Permo-Carboniferous pole position of Europe. For the Inde Syncline post-Asturian rotational deformations can be excluded. Should a N–S transport of this part of the Dinant Nappe have not occurred, the declination pattern indicates that the western part of the Dinant Nappe is transported further north than the eastern part. VGPs of the Dinant Synclinorium which deviate from the Permo-Carboniferous reference direction are due to (Asturian) syntectonic remagnetizations and subsequent 10 to 20° (maximum 40°) dextral shear deformations. These deformations can be correlated with the northwards overthrusting of the Dinant Nappe and are associated with faults at the eastern border of the Dinant Synclinorium and in the Maas valley.

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